Branded Version

Branded Product Documentation

AI-Powered Restaurant Financial Management SaaS Platform

Professional Project Report | Final Year Major Project | Startup Prototype Edition

Architecture
MERN + AI Layer
Business Scope
Finance + Inventory + Analytics
Delivery Model
Multi-tenant SaaS

1. Project Overview

This platform is a unified restaurant operations and financial intelligence system designed to replace fragmented processes with a connected SaaS workflow. It captures every transaction, expense, and inventory movement in one place and converts raw records into actionable analytics.

Problem solved: data fragmentation, manual errors, delayed decisions, and weak inventory visibility.

Core Objectives

- Track revenue and orders in real time
- Manage expenses with structured categorization
- Maintain ingredient-level inventory accuracy
- Generate management reports instantly
- Deliver AI-based forecasting and advisory insights

2. Architecture and Stack

Flow: React Dashboard -> Express API -> Business Services -> MongoDB -> AI Processing Layer

- **Frontend:** React/Next.js, Tailwind CSS, Recharts
- **Backend:** Node.js + Express.js
- **Database:** MongoDB
- **AI:** OCR + LLM + forecasting models

3. Role-Based Access (RBAC)

- **Admin:** tenant management, user control, system configuration
- **Manager:** inventory, procurement, approvals, analytics review
- **Cashier:** order creation, billing, payment recording

4. Functional Modules

Orders and POS

Table-based ordering, invoice generation, and payment settlement with audit trail.

Expenses

Receipt uploads, vendor tagging, category mapping, and correction workflow.

Inventory

Ingredient stock tracking with low-stock alerts and consumption deduction from dish sales.

Reports

Revenue, cost, profit margin, and item-level performance dashboards for decision-making.

5. AI Integration

Smart Expense Categorization

OCR extracts receipt text, AI classifies line items, and accounting categories are populated automatically.

Sales Forecasting

Historical order patterns are analyzed to predict next-week demand, peak hours, and high-volume dishes.

Business Insight Engine

AI flags margin pressure and cost anomalies, helping teams react quickly.

AI Chat Assistant

Natural language queries such as "What was today's net profit?" return direct answers from business data.

6. API Design Summary

- /api/auth/* for login, registration, and profile
- /api/orders/* for order operations
- /api/expenses/* for expense lifecycle
- /api/inventory/* for stock tracking
- /api/reports/* for financial analytics
- /api/ai/* for categorization, prediction, and chatbot

7. 10-Week Execution Plan

Research -> UI design -> backend setup -> auth -> orders -> inventory/expenses -> frontend integration -> AI integration -> testing -> deployment/documentation.

8. Startup-Ready Enhancements

- Multi-restaurant tenancy
- Cloud deployment (AWS/Vercel/DigitalOcean)
- Subscription billing via Stripe/Razorpay
- Automated weekly email reports
- Security hardening and audit logs

9. Outcome

This project demonstrates full-stack delivery, product architecture, and practical AI application. It is suitable for final-year evaluation and scalable enough to evolve into a startup MVP.

Prepared documentation variant: Branded (Logo + Theme Colors)

Viva-Focused Short Version

Viva-Focused Short Documentation

Project: AI-Powered Restaurant Financial Management SaaS Platform

One-line idea: A MERN-based SaaS platform that unifies POS, expenses, inventory, and analytics, enhanced with AI for forecasting and expense intelligence.

1. Problem and Need

- Restaurants use disconnected tools (POS + Excel + paper receipts).
- Leads to accounting errors and no real-time visibility.
- This platform centralizes all financial workflows.

2. Objectives

- Real-time sales and expense tracking
- Accurate inventory monitoring
- Automated financial reporting
- AI-driven predictions and insights

3. System Architecture

Frontend: React dashboard

Backend: Node.js + Express APIs

Database: MongoDB

AI Layer: OCR + OpenAI + forecasting logic

4. Role-Based Access

- **Admin:** complete control and tenant management
- **Manager:** expense, inventory, analytics oversight
- **Cashier:** order and billing operations

5. Core APIs (Sample)

- `POST /api/auth/login`
- `POST /api/orders/create`
- `POST /api/expenses/add`
- `POST /api/ai/predict-sales`

6. AI Highlights

- Receipt-to-category automation
- Next-week sales prediction
- Profit/margin pattern insights

- Natural-language business assistant

7. Timeline

10-week phased plan covering architecture, module build, AI integration, testing, and deployment.

8. Viva Conclusion

This project is technically strong because it combines full-stack engineering, SaaS design, data-driven analytics, and practical AI integration for a real business domain.

PPT-Ready Version

Slide 1

AI-Powered Restaurant Financial Management SaaS

Final Year Major Project + Startup Prototype

- Domain: Restaurant Operations and Finance
- Approach: MERN + AI + SaaS Multi-Tenant Design
- Goal: Real-time financial visibility and smarter decision-making

Slide 2

Problem Statement

- Fragmented systems: POS, spreadsheets, and manual inventory
- No single source of financial truth
- Manual errors and delayed reporting
- Weak cost control and stock planning

Slide 3

Proposed Solution

- Unified dashboard for sales, expense, and inventory
- Automated financial reports and KPIs
- Role-based access for Admin, Manager, Cashier
- AI assistant for forecasting and insights

Slide 4

Architecture

Frontend
React/Next.js dashboard with charts and workflows

Backend
Node.js + Express REST APIs

Database
MongoDB for transactions and tenant data

AI Layer
OCR + LLM + prediction services

Slide 5

Core Modules

- Orders & POS billing
- Expense management with receipt upload
- Inventory tracking and low-stock alerts
- Reports: revenue, profit, top dishes
- AI chat for business Q&A

Slide 6

AI Features

- Smart expense categorization from receipts
- Sales prediction for demand planning
- Margin and anomaly insights
- Natural-language analytics assistant

Slide 7

10-Week Development Plan

- W1-W2: Requirements and UI design
- W3-W6: Backend + auth + core modules
- W7-W8: Frontend integration + AI
- W9-W10: Testing, deployment, documentation

Slide 8

Business and Academic Value

- Strong final-year demo with real-world use case
- Demonstrates full-stack and AI capability
- MVP-ready architecture for startup scaling
- Future scope: subscription billing and multi-branch analytics